Bayesian Network for Student Exam Score Prediction

Enrique Such Andreu and David Valls Lozano
Master's Degree in Artificial Intelligence, University of Bologna
{ enrique.suchandreu, david.vallslozano}@studio.unibo.it

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Abstract

This project develops and analyzes a Bayesian network built from a *Student Exam Scores* dataset. Two models were implemented: an expert-driven manually constructed network and another structure learned automatically from data. Both models are compared in terms of their behavior under probabilistic inference, and several queries are carried out to evaluate performance and interpretability to realize which variables have more impact on the exam results.

Introduction

Domain

This project focuses on probabilistic modeling applied to academic performance. The objective is to represent the relationships between the variables that influence the exam results, including study habits, prior knowledge, and sleep patterns. Bayesian networks naturally model causal dependencies and enable inference over unobserved variables

Aim

The main goals of the project are:

- Build a Bayesian network based on the *Student Exam Scores* dataset.
- Compare an expert-defined structure with a structure learned automatically.
- Study the effect of discretization on the behavior of the model.
- Perform inference queries that examine how the evidence affects the exam score.

Method

The steps followed were the following:

1. Import libraries, data loading and exploratory analysis.
2. Selection of relevant variables (`Hours_Studied`, `Sleep_Hours`, `Attendance_Percent`, `Previous_Scores`, `Exam_Score`).
3. Discretization of continuous variables using manually chosen thresholds.
4. Manual construction of an expert-driven network.
5. Automatic learning of a structure using the Hill-Climbing algorithm.
6. Parameter estimation via Maximum Likelihood Estimation.
7. Execution and comparison of inference queries across both models.

Results

The analysis showed:

- The expert model provides more conservative updates.
- The learned structure reacts more strongly to evidence as a result of higher connectivity.
- Predictions match intuitive expectations: more study hours and better prior scores increase the likelihood of achieving a high exam grade.

Model

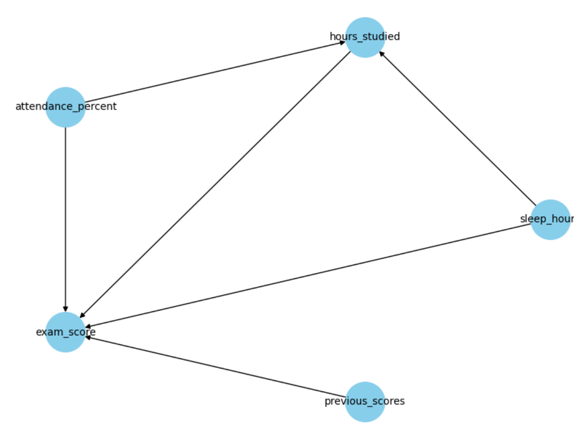

Figure 1: Expert-driven Bayesian network.

The network models the main factors affecting student's exam performance. The variables used are **Hours_Studied**, **Previous_Scores**, **Sleep_Hours**, and **Attendance_Percent**, all discretized into three levels (low/medium/high) and the target variable, **Exam_Score**, discretized into levels (fail/pass/excellent).

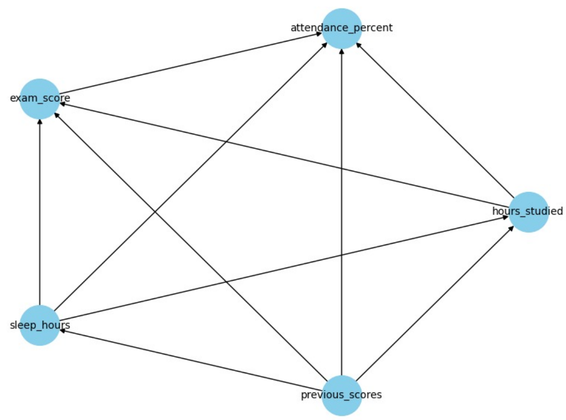

Figure 2: Data-driven Bayesian network.

Conditional distributions

All CPTs were estimated from the data using Maximum Likelihood Estimation. `Exam_Score` is conditioned on the three remaining variables; the others follow their empirical distributions.

Model construction

Two structures were created: (1) an expert-defined model assuming that all three factors directly affect the exam score; (2) a data-driven structure obtained through a score-based search. Only essential details are reported here; extended tables are provided in the notebook.

Analysis

Experimental setup

We take measures to compare the expert-driven model and the data-driven model.
The following inference queries were executed on the expert-driven model:

- $P(Exam\_Score = high \mid Hours\_Studied)$
- $P(Exam\_Score = high \mid Sleep\_Hours)$
- $P(Exam\_Score = high \mid Attendance\_Percent)$
- $P(Exam\_Score = high \mid Previous\_Scores)$
- Model variant: we compared the expert-defined model with and without the `Sleep_Hours` node to assess its contribution.

Results

The analysis showed:

- The automatic learning process suffered from overfitting, capturing noise within the dataset and detecting spurious correlations rather than true causal relationships.
- The expert-defined model is not only more interpretable, but also has a lower BIC, indicating a better compromise between fit and complexity. Therefore, it is the most appropriate model to represent student performance in a causal and understandable way.
- The model without `Sleep_Hours` caused only minor changes, confirming that this variable has a modest but stabilizing effect.
- According to the probabilities of `Exam_Score`, obtained in the different queries, it is clear that the most influential variables are `Hours_Studied` and `Previous_Scores`.

Conclusion

In conclusion, in this type of problem with few variables and clear causal relationships, automatically learned models are not very useful because they tend to detect false patterns, generate structures that are difficult to interpret, and do not reflect the logic of the domain. Furthermore, our results are conditioned because the original dataset had very low scores and we had to modify them in order to discretize them correctly, which further reduces the reliability of machine learning. Therefore, the manually defined model is more consistent, interpretable, and suitable for this analysis.

Links to external resources

Kaggle Dataset: Student Academic Performance Trends